



User Guide Manual

Mark Jeryl H. Omandam

BSIT-4

Table of Contents

What is Mikrotek?	3
Winbox	4
7 Features of Mikrotik	5
Feature 1: DHCP	5
Feature 2: NTP	10
Feature 3: Specific Device Banned	12
Feature 4: IP Reservation	18
Feature 5: Limit Bandwidth	22
Feature 6: DNS Hostname Resolver	26
Feature 7: Kid Control	30

What is Mikrotik?

MikroTik is a powerful and affordable networking platform widely used for routing, switching, security, and network management. Its core operating system, RouterOS, provides enterprise-level features such as DHCP services, NAT, firewall filtering, bandwidth control, VPN, hotspot, and advanced routing protocols. Because of its flexibility, MikroTik routers are commonly deployed in homes, businesses, schools, and even ISP-grade networks. The system is highly configurable through Winbox, WebFig, CLI, and API, making it suitable for both beginners and professional network engineers.

As a Unified Threat Management (UTM) device, MikroTik enhances network security by integrating multiple protective features firewall rules, ARP protection, DNS control, and traffic management into a single hardware or virtual appliance. This allows it to function not only as a router, but also as a security gateway that filters threats, stabilizes LAN communication, and optimizes network performance. Its versatility makes MikroTik ideal for labs, virtual environments like GNS3, and production networks that require strong control over connected devices and internet traffic.

Winbox

WinBox (64bit) v3.41 (Addresses)

File Tools

Connect To: 192.168.1.250 ☒ Keep Password

Login: admin ☐ Open In New Window

Password: **** ☒ Auto Reconnect

Add/Set Connect To RoMON Connect

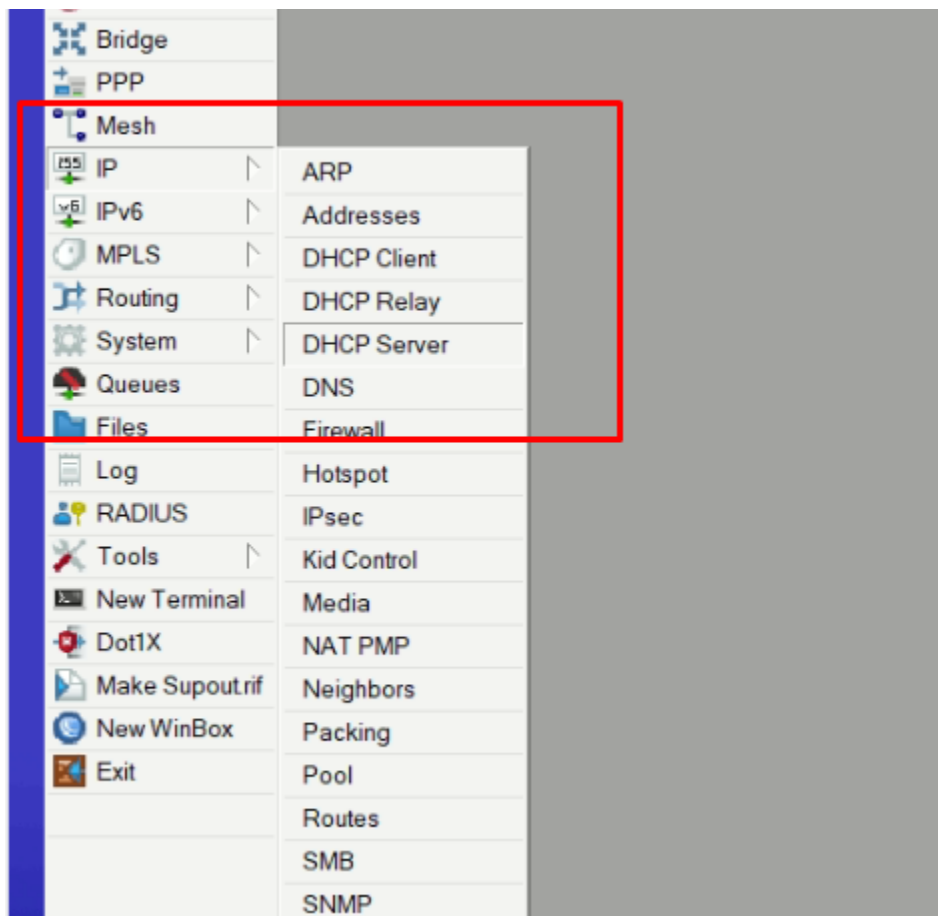
Managed Neighbors

Set Master Password Find all

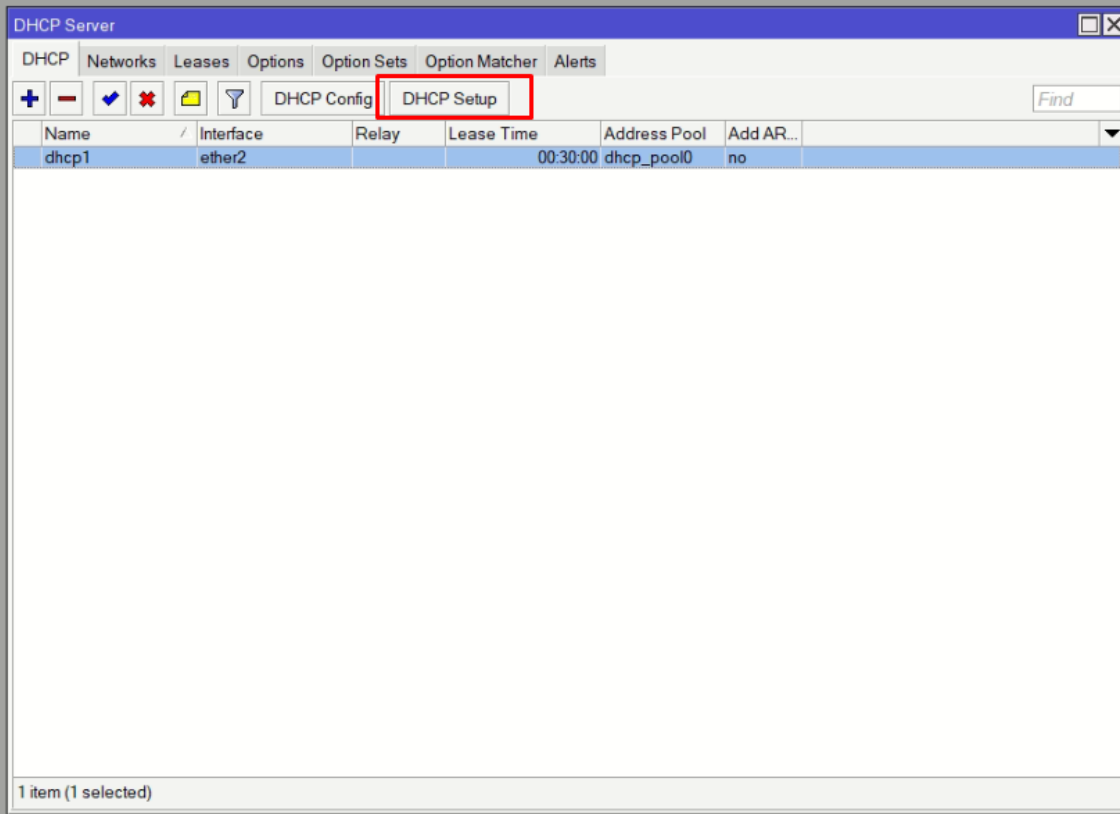
Address	User
192.168.1.250	admin

1 item

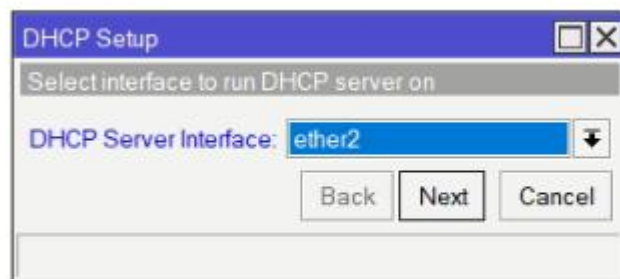
1.Feature: DHCP



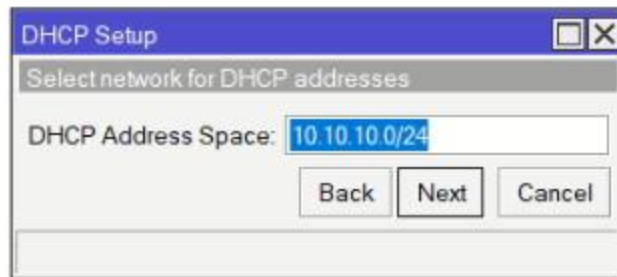
Go to IP → DHCP Server.



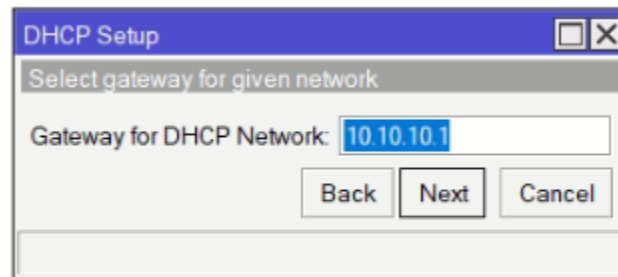
Click DHCP Setup.



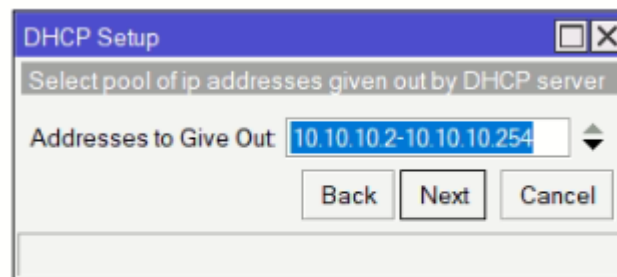
Select ether2 (your LAN interface).



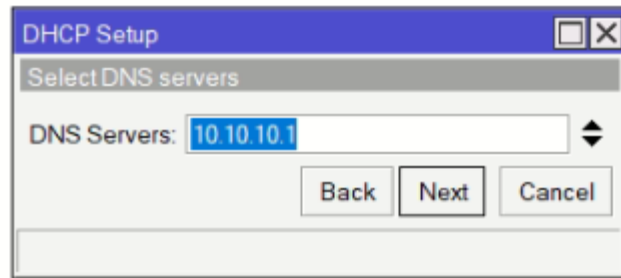
Set DHCP Address Space



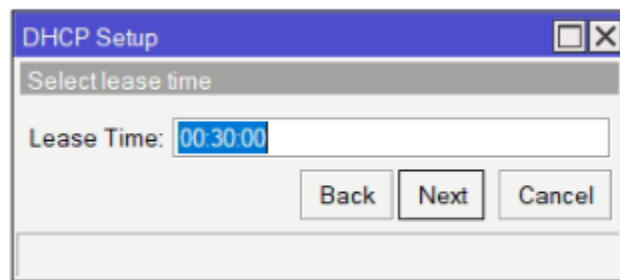
Set Gateway



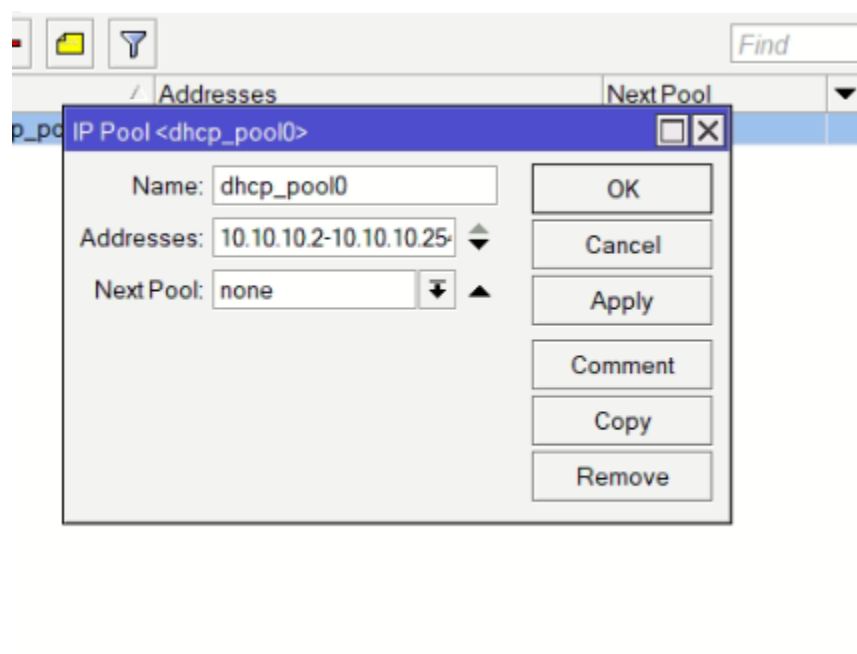
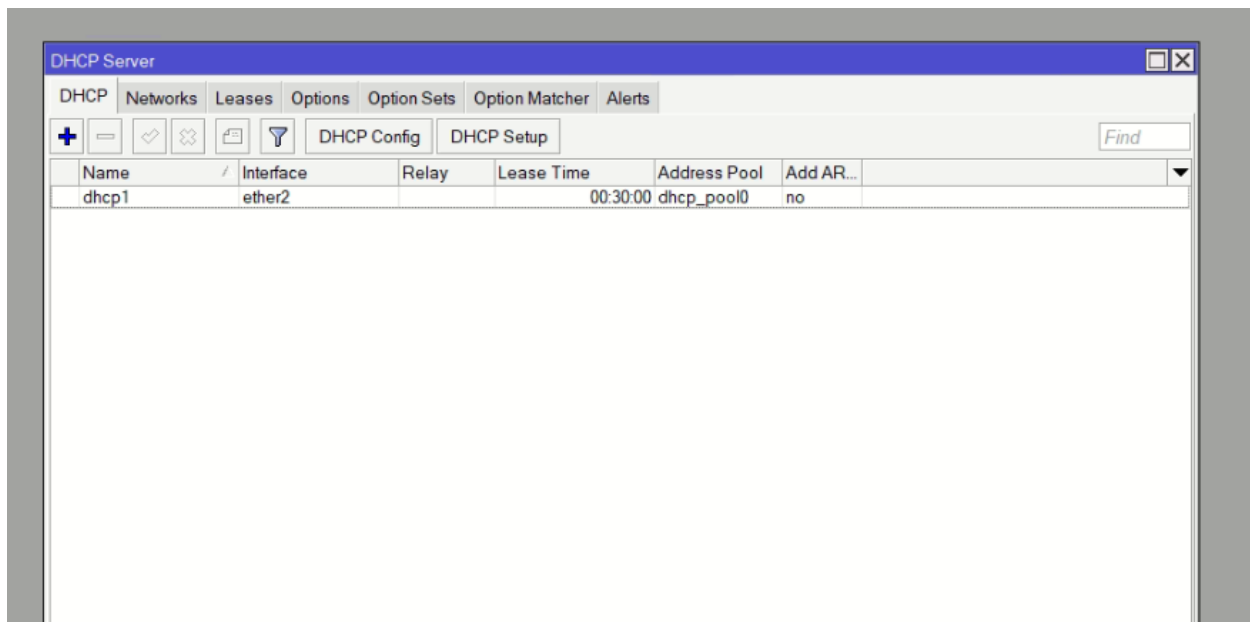
Set Address Pool



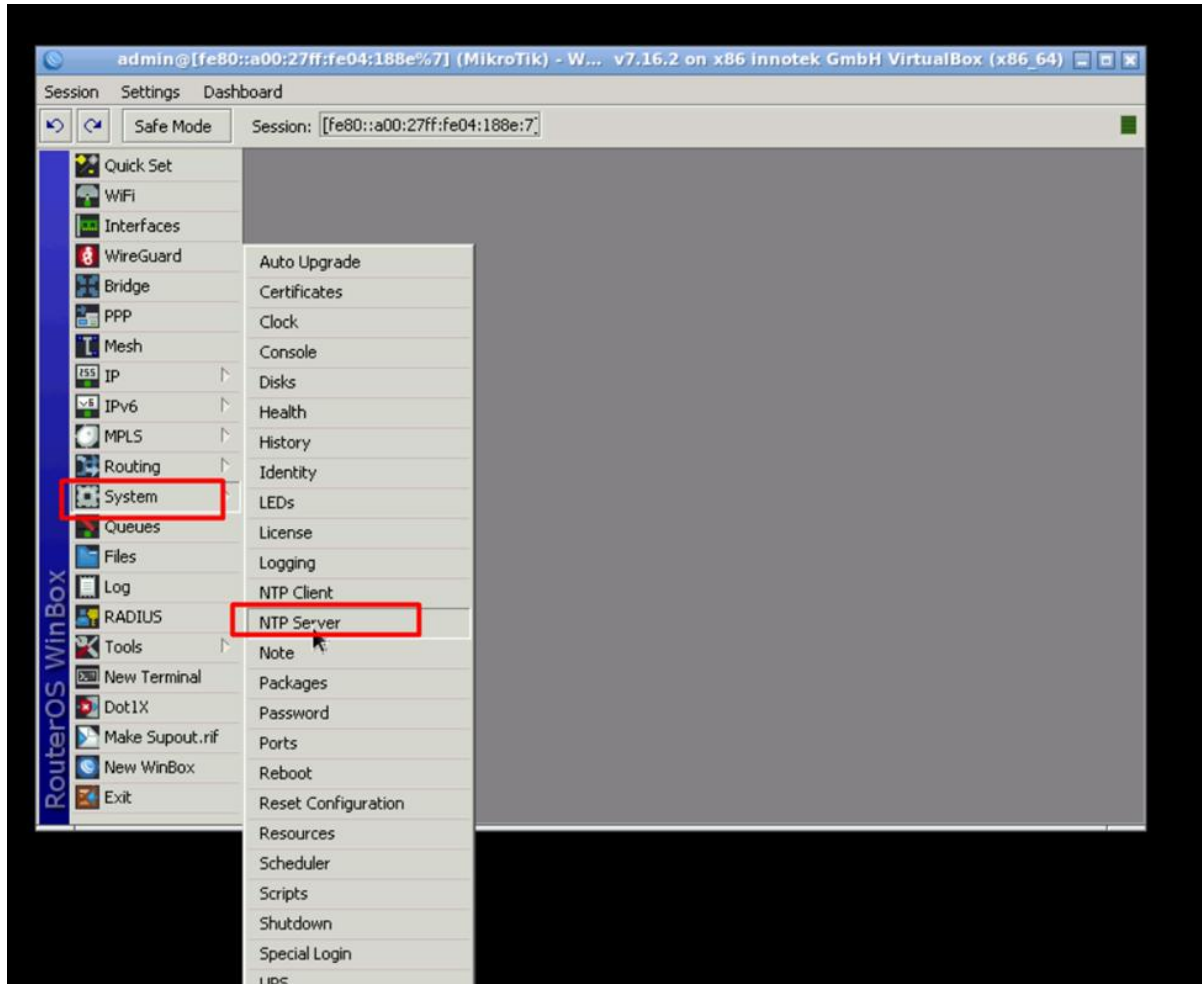
Set DNS



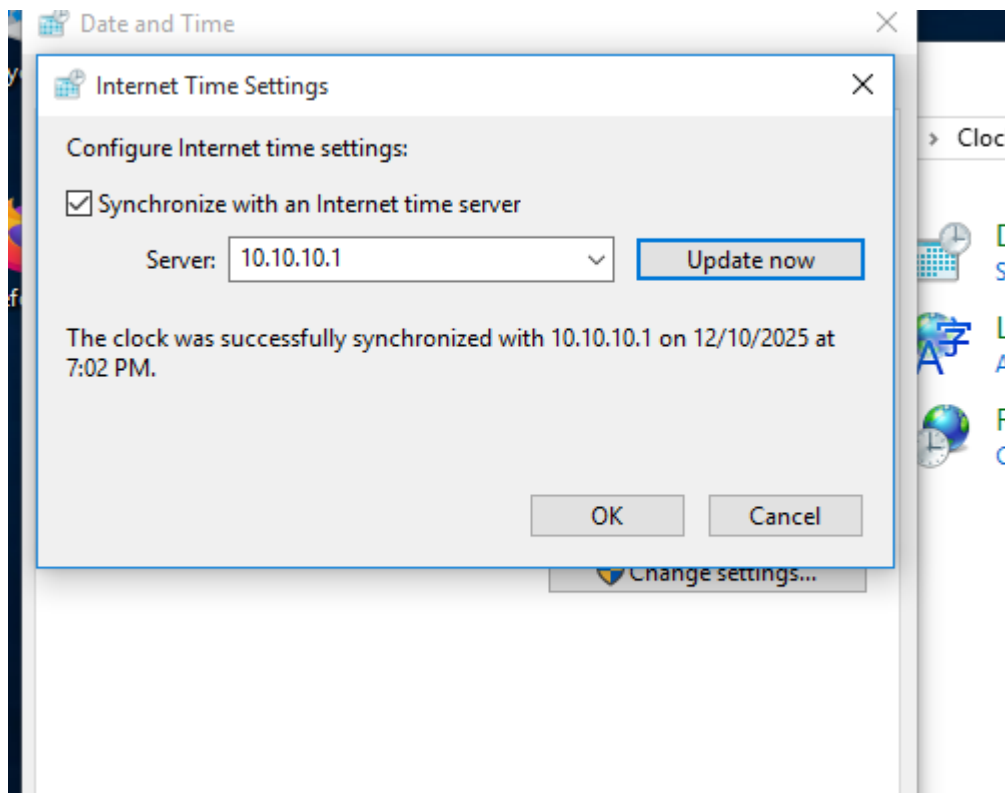
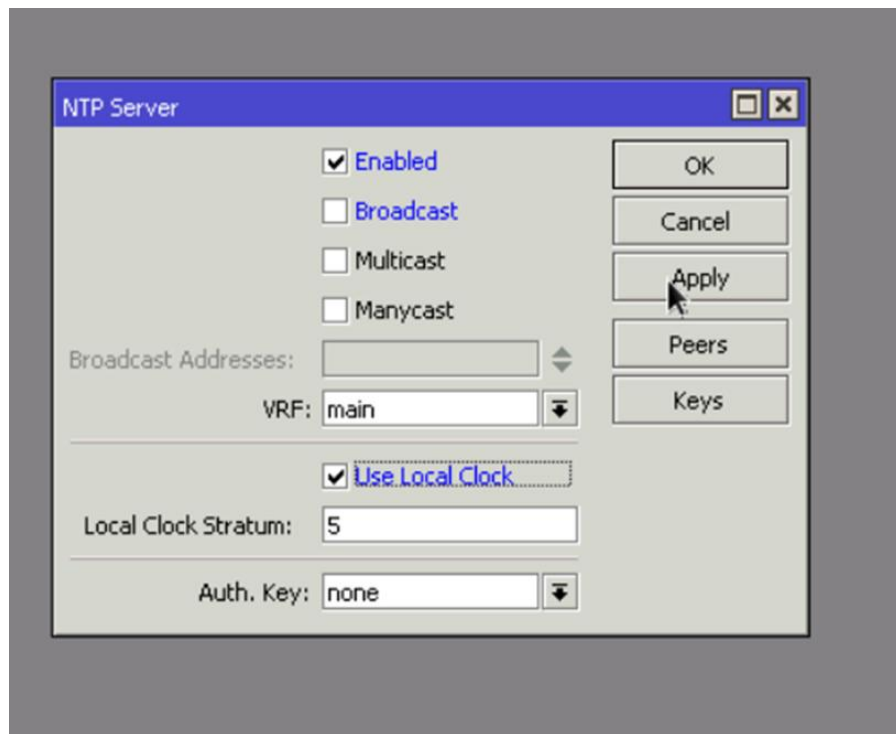
Finish the setup.



2. Feature: NTP (Network Time Protocol)

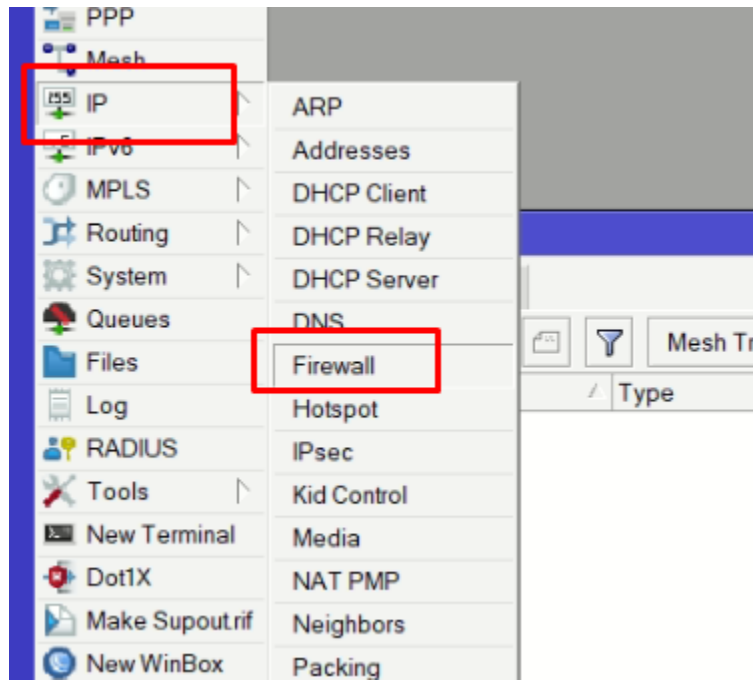


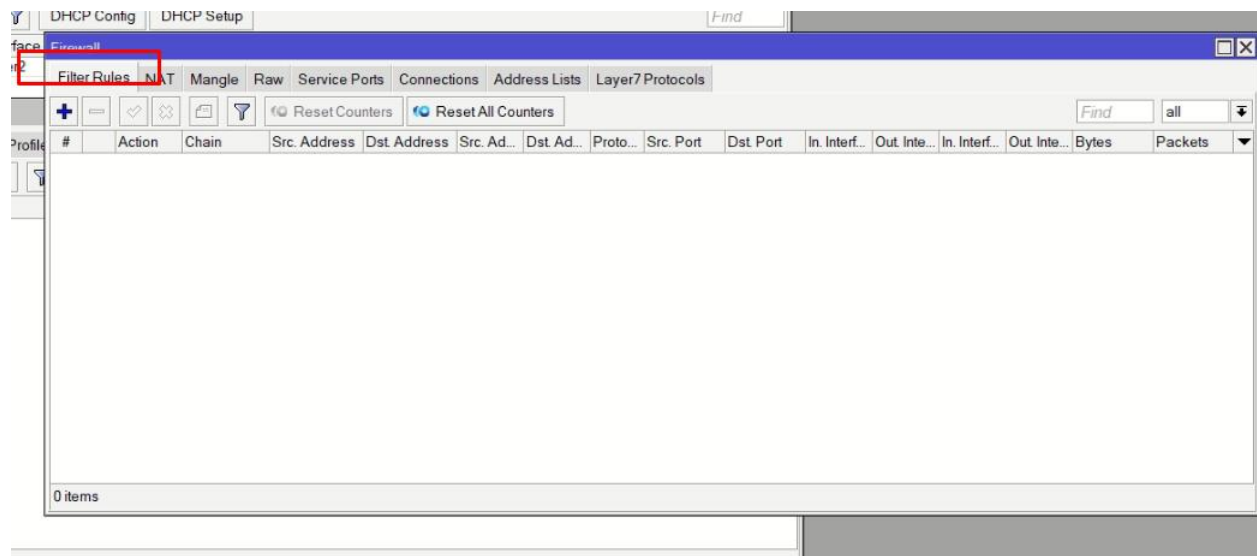
Go to **System** → **NTP SERVER**



3. Feature: Specific Device Ping Banned

Go to IP → Firewall → Filter Rules.





General tab:

- Chain: **forward**
- Src. Address: *IP of device you want to block*
- Protocol: **icmp**

The screenshot shows the 'Firewall Rule <10.10.10.30>' configuration window. The 'General' tab is selected, showing the following settings:

- Chain: **forward**
- Src. Address: (empty)
- Dst. Address: ☐ 10.10.10.30
- Src. Address List: ☐ (empty)
- Dst. Address List: (empty)
- Protocol: ☐ **1 (icmp)**
- Src. Port: (empty)
- Dst. Port: (empty)
- Any. Port: (empty)
- In. Interface: (empty)
- Out. Interface: (empty)
- In. Interface List: (empty)
- Out. Interface List: (empty)
- Packet Mark: (empty)
- Connection Mark: (empty)
- Routing Mark: (empty)
- Connection Type: (empty)
- Connection State: ☐ invalid ☒ established ☒ related ☐ new ☐ untracked
- Connection NAT State: (empty)

On the right side, there are buttons for OK, Cancel, Apply, Disable, Comment, Copy, Remove, Reset Counters, and Reset All Counters.

At the bottom left, the rule is marked as **enabled**.

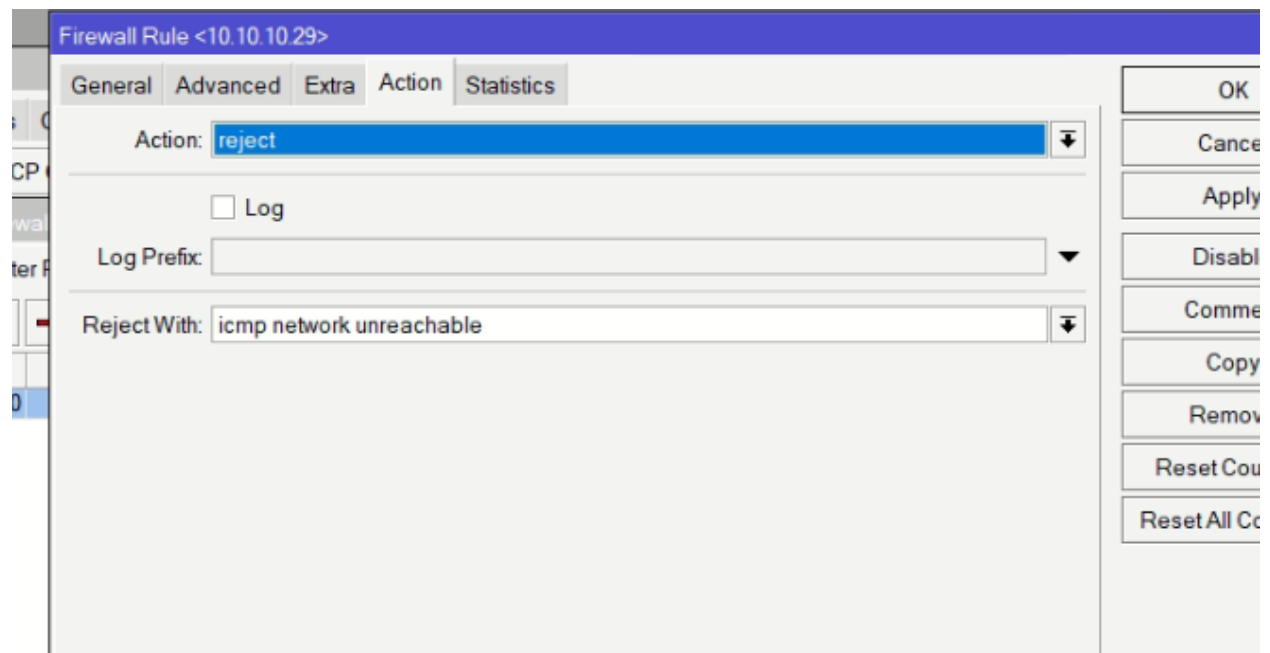
```
PC1>
PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.10.10.30/24
GATEWAY    : 10.10.10.1
DNS        : 10.10.10.1
DHCP SERVER : 10.10.10.1
DHCP LEASE  : 1089, 1800/900/1575
MAC        : 00:50:79:66:68:00
LPORT      : 20009
RHOST:PORT  : 127.0.0.1:20010
MTU        : 1500

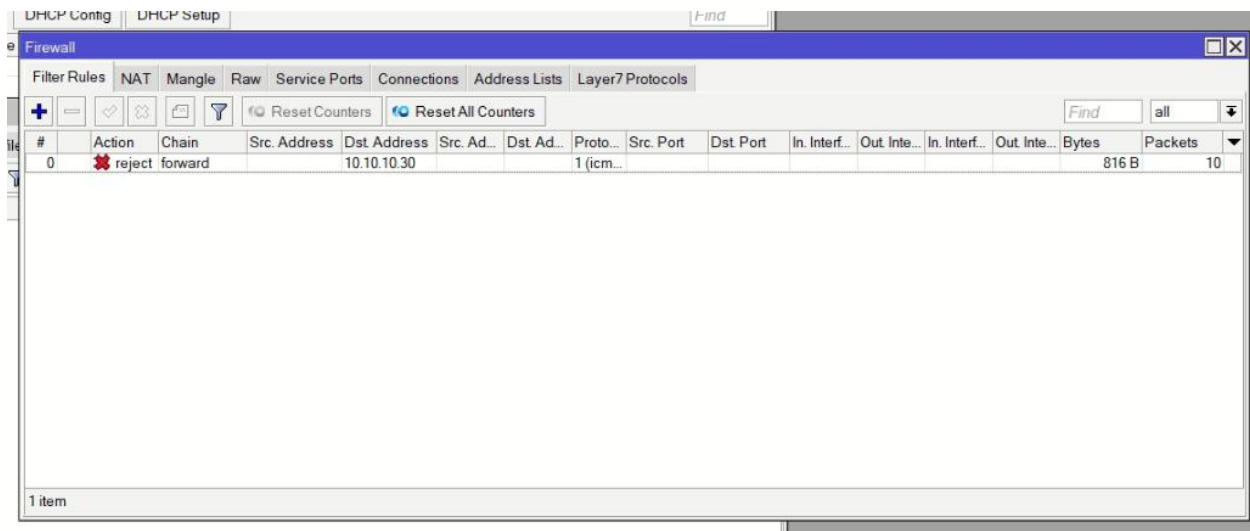
PC1> █
```

Action tab:

- **Action: drop**



Save



Before

```
PC1>
PC1> show ip
NAME       : PC1[1]
IP/MASK    : 10.10.10.30/24
GATEWAY    : 10.10.10.1
DNS        : 10.10.10.1
DHCP SERVER : 10.10.10.1
DHCP LEASE  : 1089, 1800/900/1575
MAC        : 00:50:79:66:68:00
LPORT      : 20009
RHOST:PORT  : 127.0.0.1:20010
MTU        : 1500

PC1> ping google.com
google.com resolved to 142.251.221.46

84 bytes from 142.251.221.46 icmp_seq=1 ttl=117 time=25.240 ms
84 bytes from 142.251.221.46 icmp_seq=2 ttl=117 time=24.817 ms
84 bytes from 142.251.221.46 icmp_seq=3 ttl=117 time=25.914 ms
84 bytes from 142.251.221.46 icmp_seq=4 ttl=117 time=25.848 ms
84 bytes from 142.251.221.46 icmp_seq=5 ttl=117 time=25.818 ms
```

After

```
PC1 - PuTTY
DNS      : 10.10.10.1
DHCP SERVER : 10.10.10.1
DHCP LEASE  : 1089, 1800/900/1575
MAC       : 00:50:79:66:68:00
LPORT     : 20009
RHOST:PORT : 127.0.0.1:20010
MTU       : 1500

PC1> ping google.com
google.com resolved to 142.251.221.46

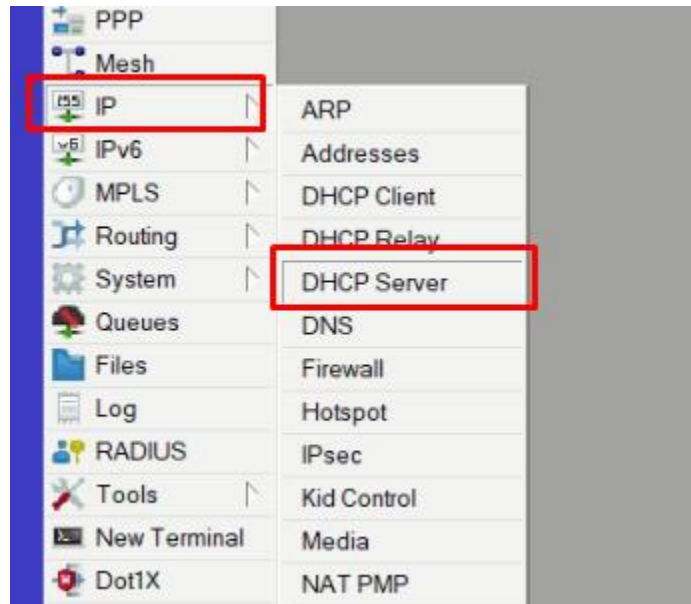
84 bytes from 142.251.221.46 icmp_seq=1 ttl=117 time=25.240 ms
84 bytes from 142.251.221.46 icmp_seq=2 ttl=117 time=24.817 ms
84 bytes from 142.251.221.46 icmp_seq=3 ttl=117 time=25.914 ms
84 bytes from 142.251.221.46 icmp_seq=4 ttl=117 time=25.848 ms
84 bytes from 142.251.221.46 icmp_seq=5 ttl=117 time=25.818 ms

PC1> ping google.com
google.com resolved to 142.251.221.46

google.com icmp_seq=1 timeout
google.com icmp_seq=2 timeout
google.com icmp_seq=3 timeout
```

4. IP Reservation

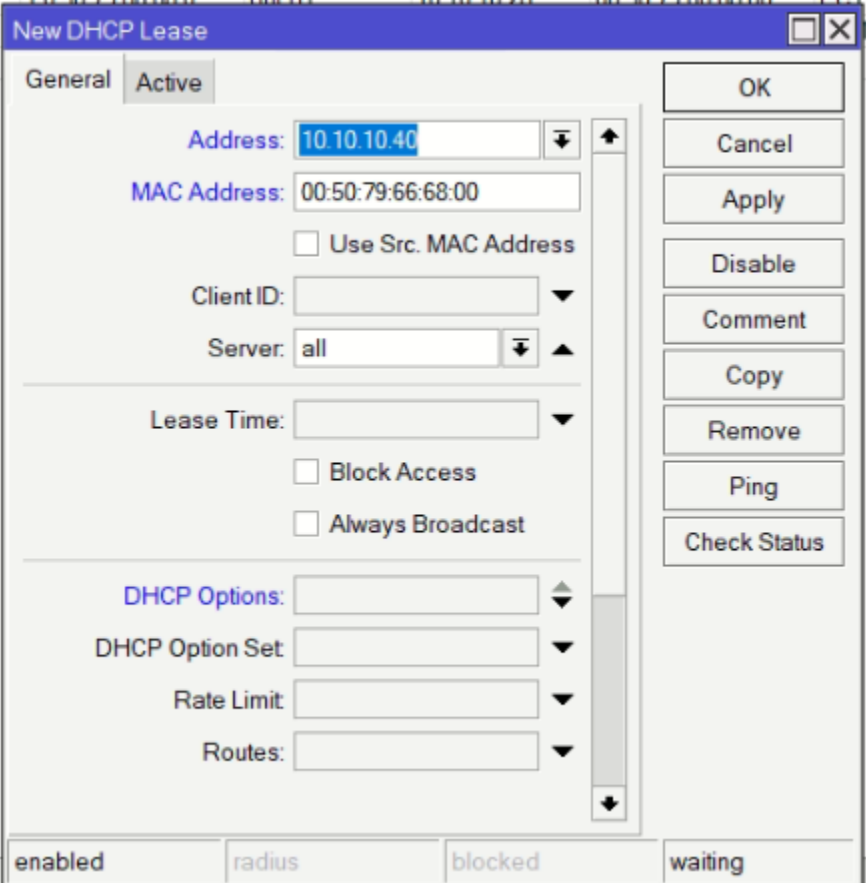
Go to IP → DHCP Server → Leases.



The screenshot shows the 'DHCP Server' window with the 'Leases' tab selected. The window contains a table of IP reservations. The table has columns for Address, MAC Address, Client ID, Server, Active Address, Active MAC Address, Active Host, and Active Class. Two reservations are listed: one for IP 10.10.10.29 and another for IP 10.10.10.30.

	Address	MAC Address	Client ID	Server	Active Address	Active MAC Address	Active Host	Active Class
D	10.10.10.29	08:00:27:01:F0:95	1:8:0:27:1:f0:95	dhcp1	10.10.10.29	08:00:27:01:F0:95	WS1	MSFT 5.0
D	10.10.10.30	00:50:79:66:68:00	1:0:50:79:66:68:0	dhcp1	10.10.10.30	00:50:79:66:68:00	PC1	

Click +
Enter Address
Mac Address



The image shows a 'New DHCP Lease' dialog box with a blue title bar and standard window controls. It has two tabs: 'General' and 'Active', with 'Active' currently selected. The 'General' tab contains several input fields and checkboxes. The 'Address' field is highlighted in blue and contains the text '10.10.10.40'. The 'MAC Address' field contains '00:50:79:66:68:00'. There is a checkbox for 'Use Src. MAC Address' which is unchecked. Below this are fields for 'Client ID' and 'Server' (set to 'all'). A 'Lease Time' field is also present. Further down are checkboxes for 'Block Access' and 'Always Broadcast', both unchecked. At the bottom of the main area are fields for 'DHCP Options', 'DHCP Option Set', 'Rate Limit', and 'Routes'. On the right side of the dialog is a vertical stack of buttons: 'OK', 'Cancel', 'Apply', 'Disable', 'Comment', 'Copy', 'Remove', 'Ping', and 'Check Status'. At the very bottom of the dialog is a status bar with four labels: 'enabled', 'radius', 'blocked', and 'waiting'.

General Active

Address: 10.10.10.40

MAC Address: 00:50:79:66:68:00

☐ Use Src. MAC Address

Client ID:

Server: all

Lease Time:

☐ Block Access

☐ Always Broadcast

DHCP Options:

DHCP Option Set:

Rate Limit:

Routes:

OK

Cancel

Apply

Disable

Comment

Copy

Remove

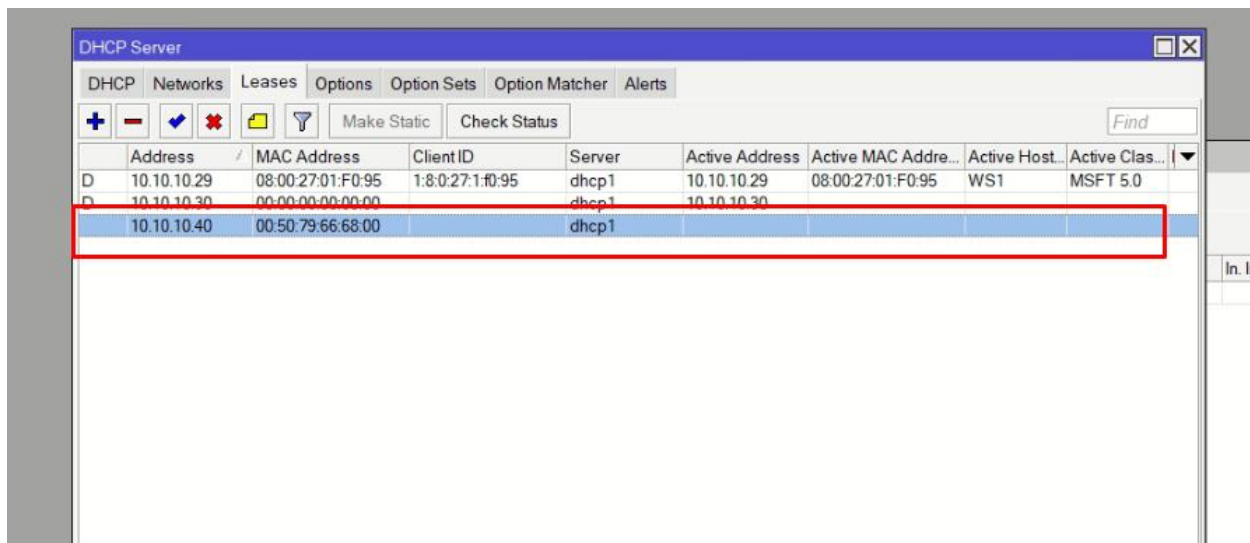
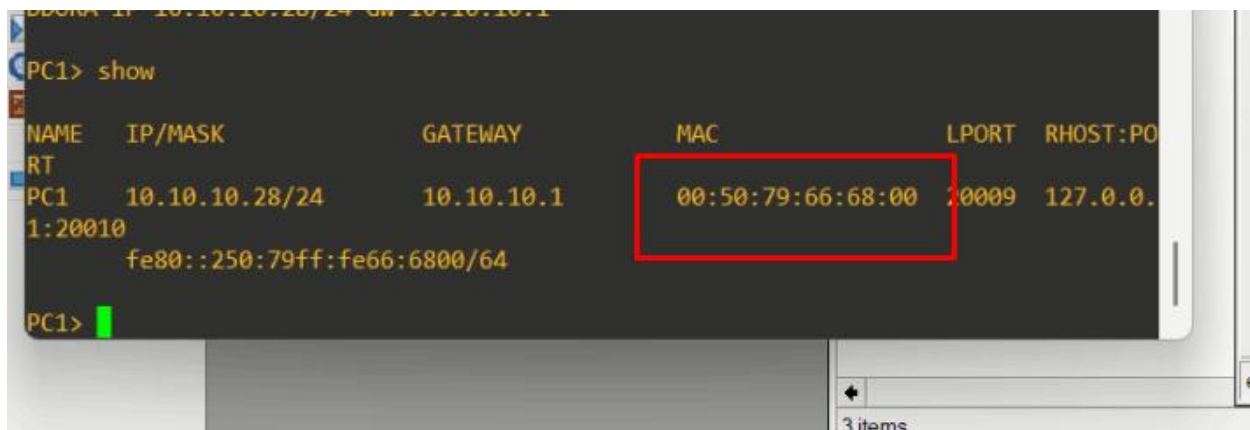
Ping

Check Status

enabled radius blocked waiting

Click apply and okay

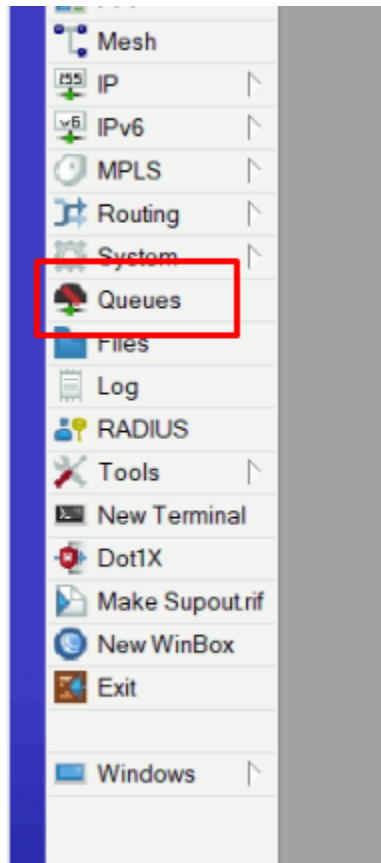
After inputting ip dhcp when you put input show ip you will see the ip you reserved



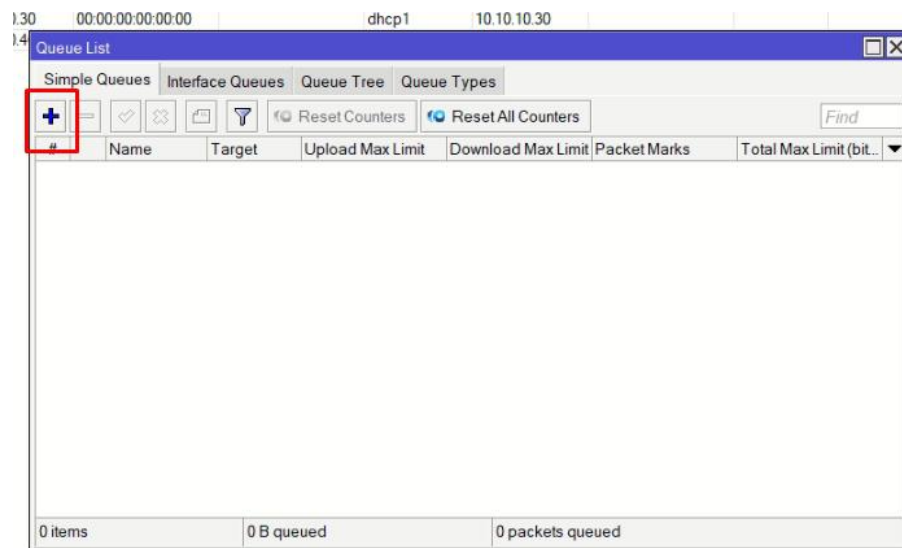
```
PC1> ip dhcp
D^[24~ORA IP 10.10.10.40/24 GW 10.10.10.1
PC1> [24~
```

5. Feature: Limit Bandwidth

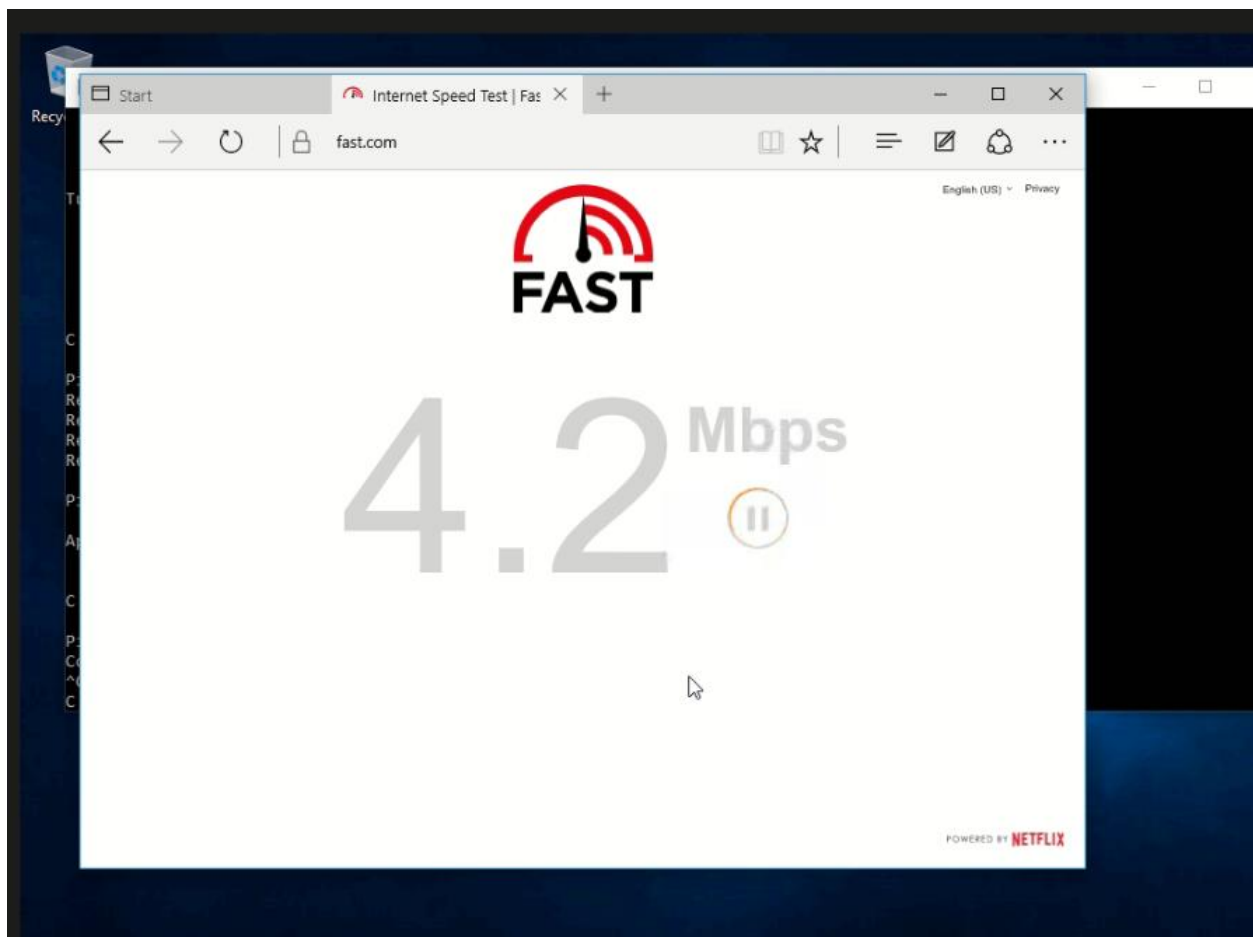
Go to Queues → Simple Queues.



Click +.

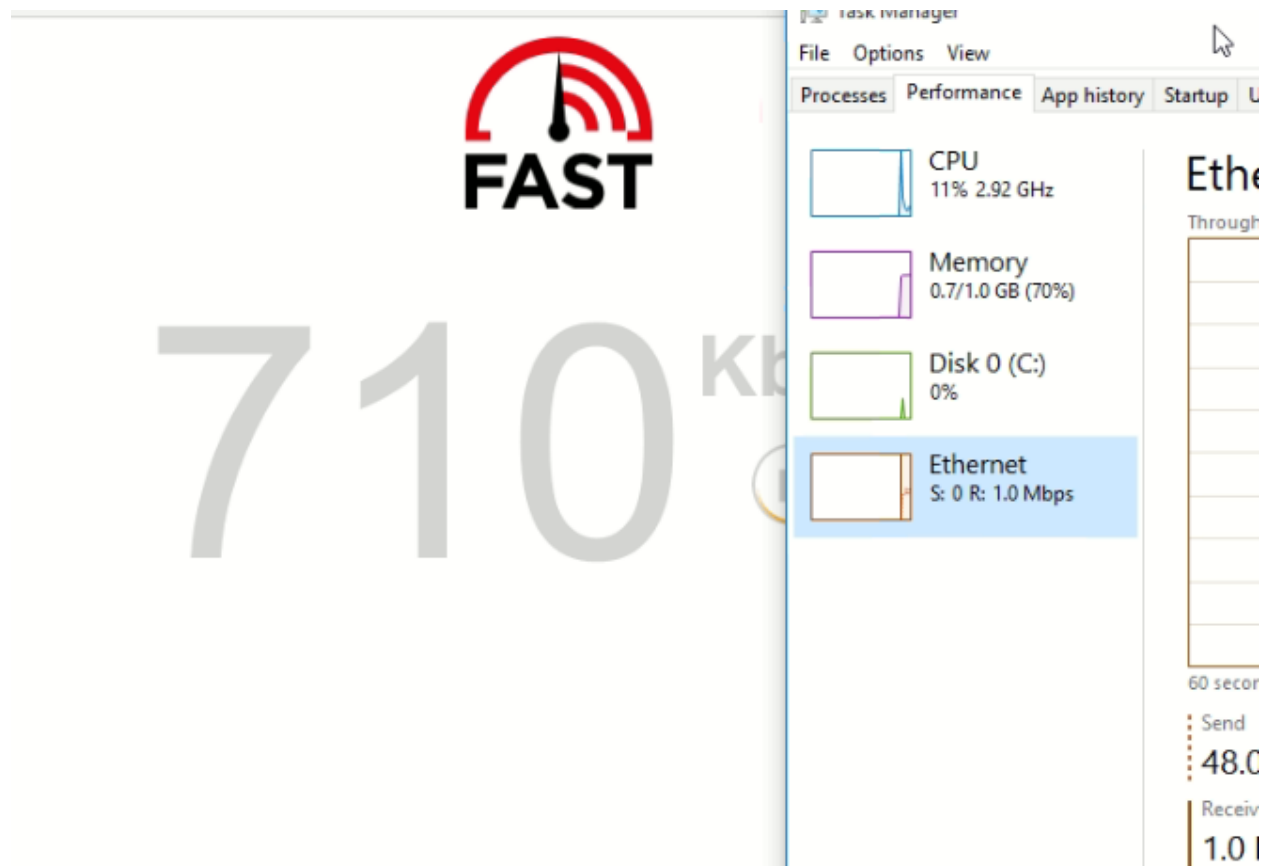


Before



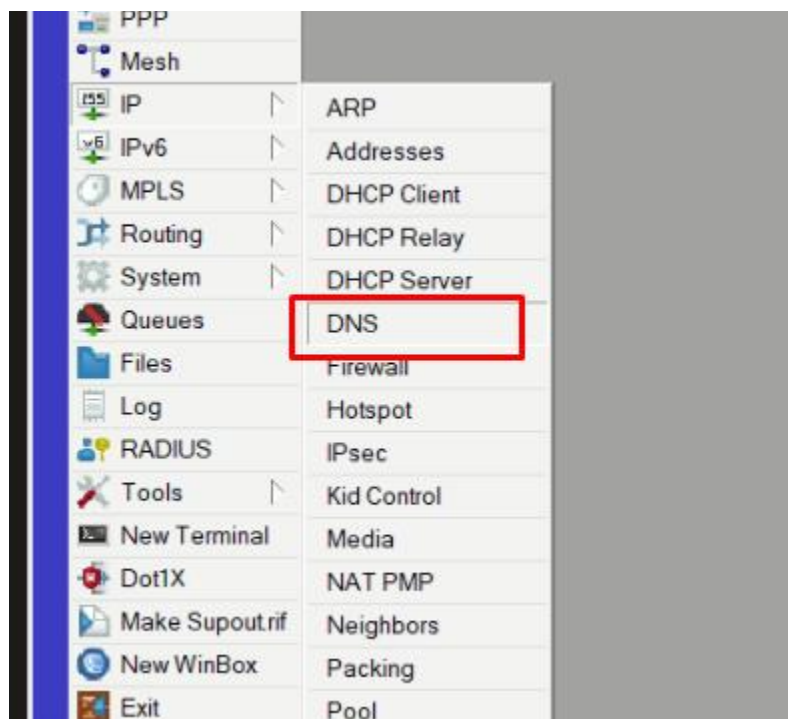
Save

24



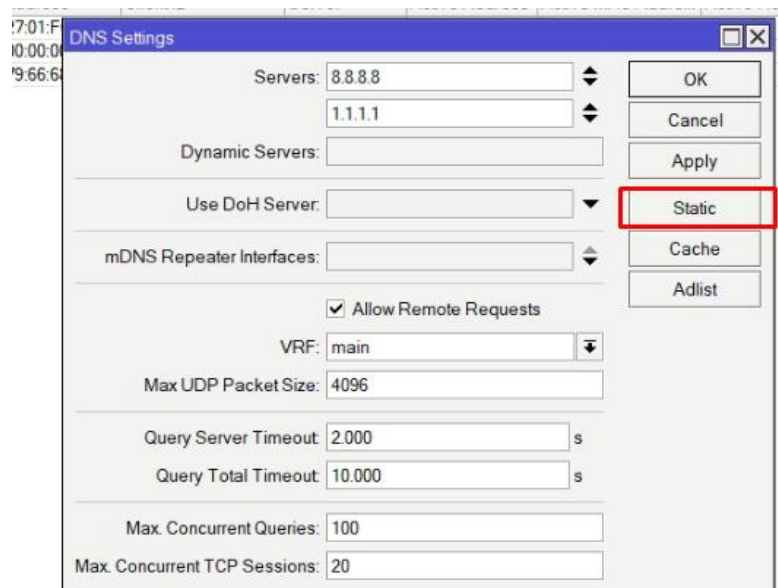
6. Feature: DNS Hostname Resolver

Go to IP → DNS.



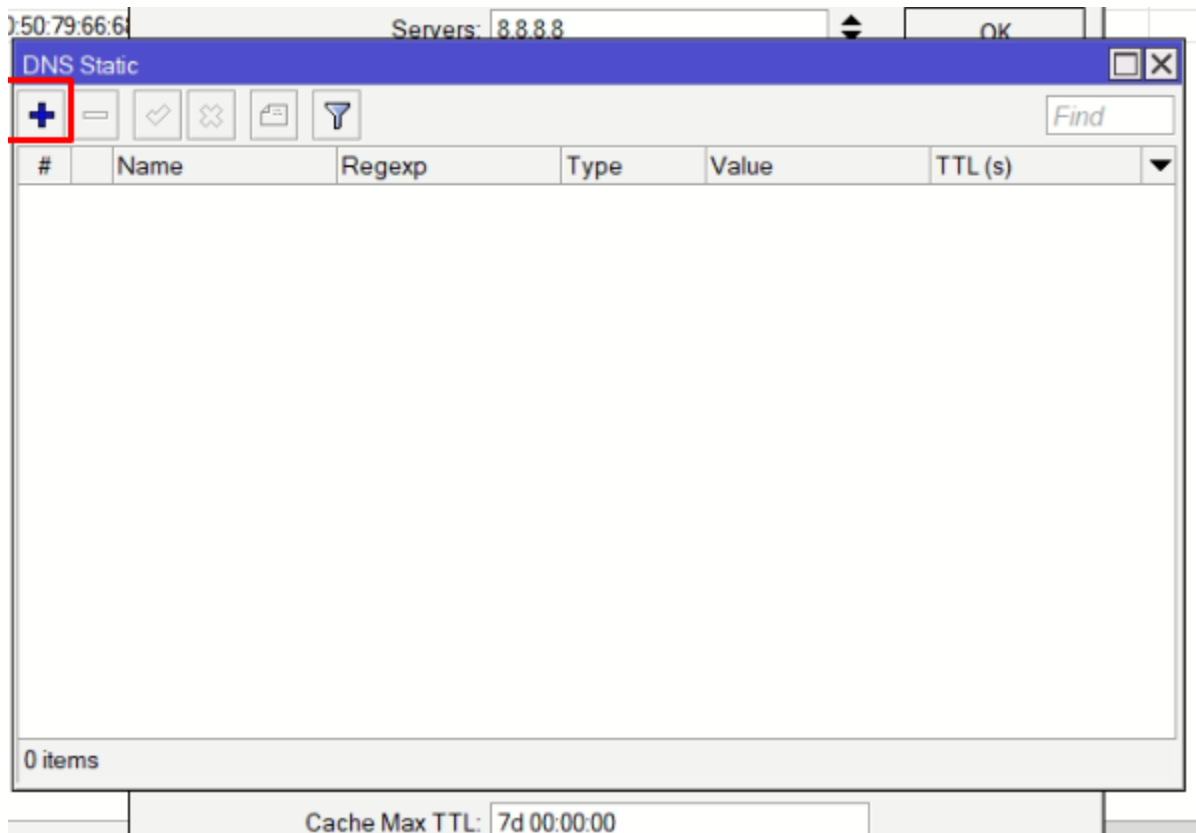
Add servers:

- 8.8.8.8
- 1.1.1.1



Add Local DNS Name

1. Go to Static tab.
2. Click +.
3. Name: *pc1.local*
4. Address: *10.10.10.10*
5. Apply → OK.



```

D- [[24-ORA IP 10.10.10.40/24 GW 10.10.10.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.10.10.40/24
GATEWAY    : 10.10.10.1
DNS        : 10.10.10.1
DHCP SERVER : 10.10.10.1
DHCP LEASE  : 1485, 1800/900/1575
MAC        : 00:50:79:66:68:00
LPORT      : 20009
RHOST:PORT  : 127.0.0.1:20010
MTU        : 1500

PC1>

```

```

C:\Users\Jeryl>ping pc1.local
Ping request could not find host pc1.local. Please check the name and try again.
C:\Users\Jeryl>

```

Send Adapter name: Ethernet

#	Name	Regexp	Type	Value	TTL (s)
0	pc1.local		A	10.10.10.40	1d 00:00:00

```
Command Prompt
C:\Users\Jeryl>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

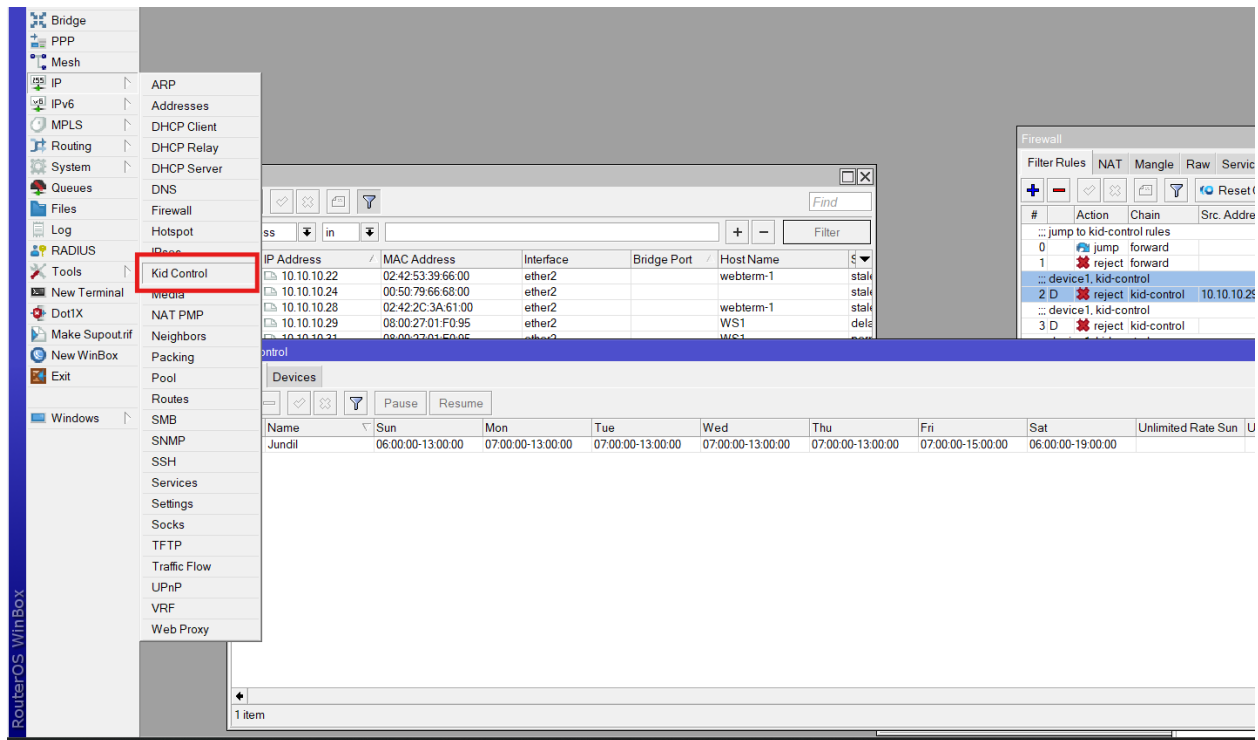
C:\Users\Jeryl>ping pc1.local

Pinging pc1.local [10.10.10.40] with 32 bytes of data:
Reply from 10.10.10.40: bytes=32 time<1ms TTL=64
Reply from 10.10.10.40: bytes=32 time<1ms TTL=64
Reply from 10.10.10.40: bytes=32 time<1ms TTL=64
Reply from 10.10.10.40: bytes=32 time<1ms TTL=64

Ping statistics for 10.10.10.40:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Jeryl>
```

7. Feature: Kid Control



Go to IP → Kid Control.

Create a profile and set allowed time slots (e.g., 7 AM - 1 PM).

Name: kid (eg.. Jundil)

Address: Enter the child's device MAC address.

Name:	Jundil	OK
Sun:	06:00:00-13:00:00	Cancel
Mon:	07:00:00-13:00:00	Apply
Tue:	07:00:00-13:00:00	Disable
Wed:	07:00:00-13:00:00	Copy
Thu:	07:00:00-13:00:00	Remove
Fri:	07:00:00-15:00:00	Pause
Sat:	06:00:00-19:00:00	Resume

Go to Devices

Click Add (+).

New Kid Control Device

Name: device1

MAC Address: 08:00:27:01:F0:95

User: Jundil

IP Address:

Rate Up/Down:

Bytes:

Activity:

OK

Cancel

Apply

Disable

Copy

Remove

Reset Counters

enabled rate limited

Click OK.

Before Kid Control

Windows 10 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

cocomelon - YouTube

www.youtube.com/results?search_query=cocomelon

cocomelon

Let's Count 5 Little Ducks! | CoComelon Nursery Rhymes and Kids Songs | Animals fo...
102M views • 7 months ago

Moonbug Kids - Animals for Kids
Mother duck is swimming with her 5 babies but one by one they all disappear...
34 chapters Five Little Ducks 2 | Sea Creature Rescue L...

Mix - Cocomelon
Cocomelon, Gooberdoo, and more
Humpty Dumpty Grocery Store | CoComelon Nursery Rhymes & Kids Songs |...
Wheels on the Bus | @CoComelon Nursery Rhymes & Kids Songs - 3:49

Cocomelon Dance
People also search for

Cocomelon

Humpty Dumpty (Grocery Store)
1.1B views • 2 years ago 2:50

Bath Song
7.3B views • 7 years ago 2:53

Baa Baa Black Sheep (Dance Party)
915M views • 3 years ago 2:45

Twinkle Twinkle Little Star (Up so High)
692M views • 3 years ago 3:16

The Lunch Song
18 views • 3 years ago 3:29

```
Microsoft Windows [Version 10.0.10240]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Users\Jeryl>ping google.com

Pinging google.com [142.251.221.14] with 32 bytes of data:
Reply from 142.251.221.14: bytes=32 time=28ms TTL=114
Reply from 142.251.221.14: bytes=32 time=33ms TTL=114
Reply from 142.251.221.14: bytes=32 time=29ms TTL=114
Reply from 142.251.221.14: bytes=32 time=28ms TTL=114

Ping statistics for 142.251.221.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 33ms, Average = 29ms

C:\Users\Jeryl>SS_
```

After Kid Control:

```
Pinging google.com [142.251.221.14] with 32 bytes of data:
Reply from 10.10.10.1: Destination net unreachable.
Reply from 10.10.10.1: Destination net unreachable.
Reply from 10.10.10.1: Destination net unreachable.
Reply from 10.10.10.1: Destination net unreachable.

Ping statistics for 142.251.221.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

